Laboratorium 4 — analiza sentymentu

Poziom 5.0 | Zadanie B | Aleksandra Gołek, Klaudia Karczmarczyk

10 662 zdania Cornell Sentence Polarity; 6 reprezentacji; 7 klasyfikatorów; 30 prób Optuny. Eksperyment: 400.9 min.

1. Cel i zakres pracy

Porównano sześć reprezentacji tekstu i siedem klasyfikatorów dla binarnej analizy sentymentu. Każdą z 42 par reprezentacja–klasyfikator strojoną metodą RandomizedSearchCV i GridSearchCV; Optuna wykonała 30 prób TPE, wspólnie wybierając reprezentację, model i jego parametry.

2. Dane i preprocessing

Cornell Sentence Polarity v1.0 zawiera 10 662 krótkie recenzje: 5 331 negatywnych (0) i 5 331 pozytywnych (1). Stratyfikowany podział 80/20, random_state=42, dał 8 529 przykładów treningowych (4 264 negatywne, 4 265 pozytywne) oraz 2 133 testowe (1 067 negatywnych, 1 066 pozytywnych).

Preprocessing: lowercase, tokenizacja, angielskie stop words, POS tagging Penn Treebank i lematyzacja WordNet. Negacje i kontrast (m.in. not, no, never, but, without oraz n't) zostały zachowane. Tagowanie, lematyzacja, Word2Vec i Doc2Vec nie korzystały z testu; parametry skalowania wektorów gęstych dopasowano wyłącznie na treningu.

3. Reprezentacje

Reprezentacja	Rodzaj i parametry	Wymiar train
Bag-of-Words	rzadka: CountVectorizer; unigramy, min_df=2, maks. 30 000 cech; macierz treningowa: 8529 × 7451	7451
Word TF-IDF	rzadka: TfidfVectorizer; n-gramy (1, 2), min_df=2, maks. 50 000 cech; macierz treningowa: 8529 × 12484	12484
Character TF-IDF	rzadka: TfidfVectorizer(analyzer='char_wb'); n-gramy znakowe (3, 5), min_df=2, maks. 30 000 cech; macierz treningowa: 8529 × 30000	30000
Word2Vec mean	gęsta, uczona lokalnie: Word2Vec Skip-gram; 100 wymiarów, okno 5, min_count=1, 20 epok; średnia wektorów słów; macierz treningowa: 8529 × 100	100
Doc2Vec	gęsta, uczona lokalnie: Doc2Vec DBOW (dm=0); 100 wymiarów, okno 5, min_count=2, 30 epok; macierz treningowa: 8529 × 100	100
GloVe mean (pretrained)	gęsta, gotowa: glove-wiki-gigaword-100; 100 wymiarów, średnia wektorów słów; macierz treningowa: 8529 × 100	100

4. Klasyfikatory i strojenie

Modele: Logistic Regression, Linear SVC, RBF SVC, K-Nearest Neighbors, Decision Tree, Random Forest i SGD Classifier. Random Search losował 2 konfiguracje na parę, Grid Search przeglądał ograniczone siatki, a Optuna przeprowadziła 30 próby TPE. W każdej metodzie użyto 2-fold StratifiedKFold z tasowaniem i random_state=42; optymalizowano F1 klasy pozytywnej. Zbiór testowy oceniono dopiero po wyborze parametrów. Każdy model stroił na wspólnej, stratyfikowanej próbce 1500 przykładów, a wybrane konfiguracje dopasowano ponownie na pełnym treningu, z wyjątkiem RBF SVC, którego końcowe dopasowanie na próbce 1500 rekordów ogranicza koszt obliczania jądra.

Zakresy hiperparametrów

Klasyfikator	Random Search	Grid Search
Logistic Regression	{"C": [0.1, 1.0, 10.0], "class_weight": [null, "balanced"]}	{"C": [0.1, 10.0], "class_weight": [null]}
Linear SVC	{"C": [0.1, 1.0, 10.0], "class_weight": [null, "balanced"]}	{"C": [0.1, 10.0], "class_weight": [null]}
RBF SVC	{"C": [0.1, 1.0, 10.0], "class_weight": [null, "balanced"], "gamma": ["scale", "auto"]}	{"C": [0.1, 10.0], "class_weight": [null], "gamma": ["scale"]}
K-Nearest Neighbors	{"n_neighbors": [3, 5, 7], "weights": ["uniform", "distance"]}	{"n_neighbors": [3, 7], "weights": ["distance"]}

Klasyfikator	Random Search	Grid Search
Decision Tree	{"criterion": ["gini", "entropy"], "max_depth": [null, 20, 50], "min_samples_leaf": [1, 2]}	{"criterion": ["gini"], "max_depth": [null, 20], "min_samples_leaf": [1]}
Random Forest	{"max_depth": [null, 30], "max_features": ["sqrt", 0.5], "min_samples_leaf": [1, 2]}	{"max_depth": [null, 30], "max_features": ["sqrt"], "min_samples_leaf": [1]}
SGD Classifier	{"alpha": [1e-05, 0.0001, 0.001], "class_weight": [null, "balanced"], "loss": ["hinge", "log_loss"]}	{"alpha": [1e-05, 0.001], "class_weight": [null], "loss": ["hinge"]}

5. Wyniki — porównanie metod

Metoda	Reprezentacja	Klasyfikator	CV F1	Accuracy	F1
Random Search	GloVe mean (pretrained)	Linear SVC	0.6822	0.7168	0.7167
Grid Search	GloVe mean (pretrained)	Linear SVC	0.6843	0.7168	0.7167
Optuna	GloVe mean (pretrained)	Linear SVC	0.6822	0.7168	0.7167

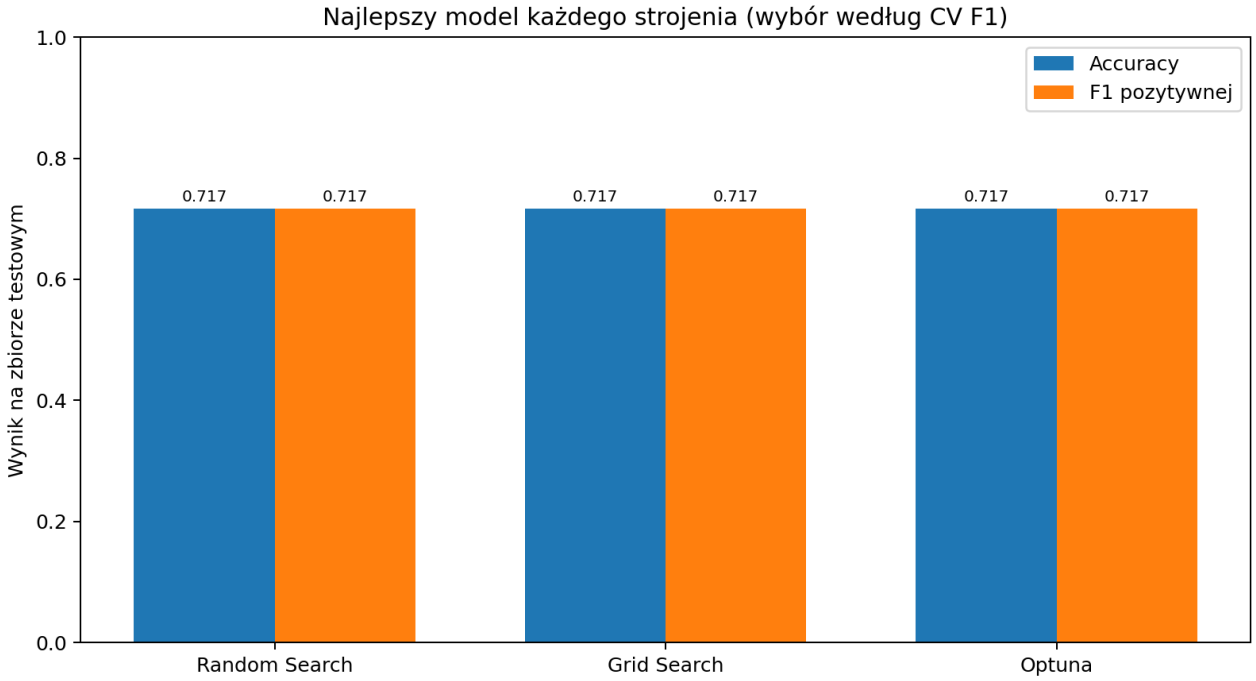
Wyniki wszystkich par reprezentacja–klasyfikator i metod strojenia

Poniższa tabela raportuje wszystkie 84 wyniki Random/Grid oraz wynik wybrany przez Optunę. Parametry wybierano na CV; metryki testowe nie wpływały na wybór.

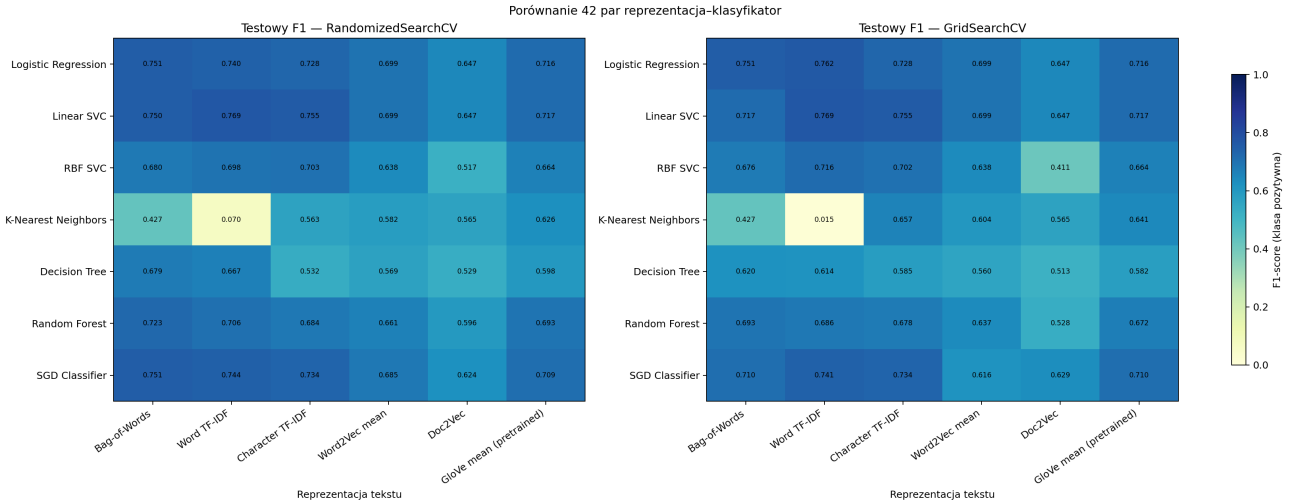
Reprezentacja	Klasyfikator	Metoda	CV F1	Accuracy	F1	Najlepsze parametry
Bag-of-Words	Logistic Regression	RandomizedSearchCV	0.6252	0.7487	0.7512	{"C": 0.1, "class_weight": null}
Bag-of-Words	Logistic Regression	GridSearchCV	0.6252	0.7487	0.7512	{"C": 0.1, "class_weight": null}
Bag-of-Words	Linear SVC	RandomizedSearchCV	0.6200	0.7478	0.7502	{"C": 0.1, "class_weight": null}
Bag-of-Words	Linear SVC	GridSearchCV	0.6220	0.7159	0.7166	{"C": 10.0, "class_weight": null}
Bag-of-Words	RBF SVC	RandomizedSearchCV	0.6727	0.6442	0.6804	{"C": 10.0, "class_weight": null, "gamma": "auto"}
Bag-of-Words	RBF SVC	GridSearchCV	0.6257	0.6728	0.6763	{"C": 10.0, "class_weight": null, "gamma": "scale"}
Bag-of-Words	K-Nearest Neighbors	RandomizedSearchCV	0.6258	0.5762	0.4271	{"n_neighbors": 3, "weights": "uniform"}
Bag-of-Words	K-Nearest Neighbors	GridSearchCV	0.6258	0.5762	0.4271	{"n_neighbors": 3, "weights": "distance"}
Bag-of-Words	Decision Tree	RandomizedSearchCV	0.5230	0.5959	0.6788	{"criterion": "entropy", "max_depth": 50, "min_samples_leaf": 1}
Bag-of-Words	Decision Tree	GridSearchCV	0.5887	0.6409	0.6197	{"criterion": "gini", "max_depth": null, "min_samples_leaf": 1}
Bag-of-Words	Random Forest	RandomizedSearchCV	0.5611	0.7271	0.7234	{"max_depth": null, "max_features": "sqrt", "min_samples_leaf": 2}
Bag-of-Words	Random Forest	GridSearchCV	0.6019	0.7121	0.6933	{"max_depth": null, "max_features": "sqrt", "min_samples_leaf": 1}
Bag-of-Words	SGD Classifier	RandomizedSearchCV	0.6219	0.7468	0.7507	{"alpha": 0.001, "class_weight": null, "loss": "log_loss"}
Bag-of-Words	SGD Classifier	GridSearchCV	0.6074	0.7164	0.7101	{"alpha": 1e-05, "class_weight": null, "loss": "hinge"}
Word TF-IDF	Logistic Regression	RandomizedSearchCV	0.6311	0.7375	0.7395	{"C": 0.1, "class_weight": null}
Word TF-IDF	Logistic Regression	GridSearchCV	0.6349	0.7595	0.7617	{"C": 10.0, "class_weight": null}
Word TF-IDF	Linear SVC	RandomizedSearchCV	0.6347	0.7665	0.7692	{"C": 0.1, "class_weight": null}
Word TF-IDF	Linear SVC	GridSearchCV	0.6347	0.7665	0.7692	{"C": 0.1, "class_weight": null}
Word TF-IDF	RBF SVC	RandomizedSearchCV	0.6344	0.6934	0.6978	{"C": 10.0, "class_weight": "balanced", "gamma": "scale"}
Word TF-IDF	RBF SVC	GridSearchCV	0.6464	0.6376	0.7163	{"C": 0.1, "class_weight": null, "gamma": "scale"}
Word TF-IDF	K-Nearest Neighbors	RandomizedSearchCV	0.6146	0.5152	0.0701	{"n_neighbors": 3, "weights": "uniform"}
Word TF-IDF	K-Nearest Neighbors	GridSearchCV	0.6235	0.5035	0.0149	{"n_neighbors": 7, "weights": "distance"}
Word TF-IDF	Decision Tree	RandomizedSearchCV	0.4960	0.5949	0.6667	{"criterion": "entropy", "max_depth": 50, "min_samples_leaf": 1}
Word TF-IDF	Decision Tree	GridSearchCV	0.5984	0.6249	0.6135	{"criterion": "gini", "max_depth": null, "min_samples_leaf": 1}
Word TF-IDF	Random Forest	RandomizedSearchCV	0.5887	0.7135	0.7064	{"max_depth": null, "max_features": "sqrt", "min_samples_leaf": 2}
Word TF-IDF	Random Forest	GridSearchCV	0.6242	0.7089	0.6856	{"max_depth": null, "max_features": "sqrt", "min_samples_leaf": 1}
Word TF-IDF	SGD Classifier	RandomizedSearchCV	0.6349	0.7412	0.7437	{"alpha": 0.001, "class_weight": null, "loss": "log_loss"}

Reprezentacja	Klasyfikator	Metoda	CV F1	Accuracy	F1	Najlepsze parametry
Word TF-IDF	SGD Classifier	GridSearchCV	0.6258	0.7342	0.7412	{"alpha": 0.001, "class_weight": null, "loss": "hinge"}
Character TF-IDF	Logistic Regression	RandomizedSearchCV	0.6518	0.7304	0.7281	{"C": 0.1, "class_weight": null}
Character TF-IDF	Logistic Regression	GridSearchCV	0.6518	0.7304	0.7281	{"C": 0.1, "class_weight": null}
Character TF-IDF	Linear SVC	RandomizedSearchCV	0.6458	0.7543	0.7551	{"C": 0.1, "class_weight": null}
Character TF-IDF	Linear SVC	GridSearchCV	0.6458	0.7543	0.7551	{"C": 0.1, "class_weight": null}
Character TF-IDF	RBF SVC	RandomizedSearchCV	0.6578	0.6554	0.7030	{"C": 10.0, "class_weight": null, "gamma": "auto"}
Character TF-IDF	RBF SVC	GridSearchCV	0.6414	0.6109	0.7017	{"C": 0.1, "class_weight": null, "gamma": "scale"}
Character TF-IDF	K-Nearest Neighbors	RandomizedSearchCV	0.5940	0.6446	0.5634	{"n_neighbors": 3, "weights": "uniform"}
Character TF-IDF	K-Nearest Neighbors	GridSearchCV	0.6089	0.6915	0.6573	{"n_neighbors": 7, "weights": "distance"}
Character TF-IDF	Decision Tree	RandomizedSearchCV	0.5256	0.5823	0.5318	{"criterion": "entropy", "max_depth": 50, "min_samples_leaf": 1}
Character TF-IDF	Decision Tree	GridSearchCV	0.5354	0.5907	0.5849	{"criterion": "gini", "max_depth": null, "min_samples_leaf": 1}
Character TF-IDF	Random Forest	RandomizedSearchCV	0.5738	0.6953	0.6842	{"max_depth": 30, "max_features": "sqrt", "min_samples_leaf": 2}
Character TF-IDF	Random Forest	GridSearchCV	0.5988	0.6892	0.6780	{"max_depth": 30, "max_features": "sqrt", "min_samples_leaf": 1}
Character TF-IDF	SGD Classifier	RandomizedSearchCV	0.6459	0.7332	0.7340	{"alpha": 0.001, "class_weight": null, "loss": "log_loss"}
Character TF-IDF	SGD Classifier	GridSearchCV	0.6353	0.7421	0.7343	{"alpha": 0.001, "class_weight": null, "loss": "hinge"}
Word2Vec mean	Logistic Regression	RandomizedSearchCV	0.6692	0.7000	0.6990	{"C": 0.1, "class_weight": null}
Word2Vec mean	Logistic Regression	GridSearchCV	0.6692	0.7000	0.6990	{"C": 0.1, "class_weight": null}
Word2Vec mean	Linear SVC	RandomizedSearchCV	0.6590	0.7000	0.6987	{"C": 0.1, "class_weight": null}
Word2Vec mean	Linear SVC	GridSearchCV	0.6590	0.7000	0.6987	{"C": 0.1, "class_weight": null}
Word2Vec mean	RBF SVC	RandomizedSearchCV	0.6315	0.6362	0.6381	{"C": 10.0, "class_weight": "balanced", "gamma": "scale"}
Word2Vec mean	RBF SVC	GridSearchCV	0.6315	0.6362	0.6381	{"C": 10.0, "class_weight": null, "gamma": "scale"}
Word2Vec mean	K-Nearest Neighbors	RandomizedSearchCV	0.5823	0.5992	0.5823	{"n_neighbors": 3, "weights": "uniform"}
Word2Vec mean	K-Nearest Neighbors	GridSearchCV	0.6042	0.6259	0.6038	{"n_neighbors": 7, "weights": "distance"}
Word2Vec mean	Decision Tree	RandomizedSearchCV	0.5616	0.5720	0.5687	{"criterion": "entropy", "max_depth": 50, "min_samples_leaf": 1}
Word2Vec mean	Decision Tree	GridSearchCV	0.5659	0.5654	0.5596	{"criterion": "gini", "max_depth": null, "min_samples_leaf": 1}
Word2Vec mean	Random Forest	RandomizedSearchCV	0.6417	0.6774	0.6607	{"max_depth": null, "max_features": "sqrt", "min_samples_leaf": 2}
Word2Vec mean	Random Forest	GridSearchCV	0.6292	0.6559	0.6374	{"max_depth": null, "max_features": "sqrt", "min_samples_leaf": 1}
Word2Vec mean	SGD Classifier	RandomizedSearchCV	0.6597	0.6873	0.6852	{"alpha": 0.001, "class_weight": null, "loss": "log_loss"}
Word2Vec mean	SGD Classifier	GridSearchCV	0.6214	0.6198	0.6162	{"alpha": 1e-05, "class_weight": null, "loss": "hinge"}
Doc2Vec	Logistic Regression	RandomizedSearchCV	0.5543	0.6357	0.6473	{"C": 0.1, "class_weight": null}
Doc2Vec	Logistic Regression	GridSearchCV	0.5543	0.6357	0.6473	{"C": 0.1, "class_weight": null}
Doc2Vec	Linear SVC	RandomizedSearchCV	0.5427	0.6357	0.6470	{"C": 0.1, "class_weight": null}
Doc2Vec	Linear SVC	GridSearchCV	0.5437	0.6357	0.6470	{"C": 10.0, "class_weight": null}
Doc2Vec	RBF SVC	RandomizedSearchCV	0.5480	0.5607	0.5173	{"C": 10.0, "class_weight": "balanced", "gamma": "scale"}
Doc2Vec	RBF SVC	GridSearchCV	0.5686	0.5331	0.4113	{"C": 0.1, "class_weight": null, "gamma": "scale"}
Doc2Vec	K-Nearest Neighbors	RandomizedSearchCV	0.5175	0.5762	0.5654	{"n_neighbors": 3, "weights": "uniform"}
Doc2Vec	K-Nearest Neighbors	GridSearchCV	0.5175	0.5762	0.5654	{"n_neighbors": 3, "weights": "distance"}
Doc2Vec	Decision Tree	RandomizedSearchCV	0.5101	0.5401	0.5286	{"criterion": "entropy", "max_depth": 50, "min_samples_leaf": 1}
Doc2Vec	Decision Tree	GridSearchCV	0.5106	0.4984	0.5128	{"criterion": "gini", "max_depth": null, "min_samples_leaf": 1}
Doc2Vec	Random Forest	RandomizedSearchCV	0.5268	0.5874	0.5956	{"max_depth": null, "max_features": "sqrt", "min_samples_leaf": 2}
Doc2Vec	Random Forest	GridSearchCV	0.5256	0.5438	0.5284	{"max_depth": null, "max_features": "sqrt", "min_samples_leaf": 1}
Doc2Vec	SGD Classifier	RandomizedSearchCV	0.5588	0.6212	0.6242	{"alpha": 0.001, "class_weight": "balanced", "loss": "hinge"}
Doc2Vec	SGD Classifier	GridSearchCV	0.5588	0.6254	0.6292	{"alpha": 0.001, "class_weight": null, "loss": "hinge"}

Reprezentacja	Klasyfikator	Metoda	CV F1	Accuracy	F1	Najlepsze parametry
GloVe mean (pretrained)	Logistic Regression	RandomizedSearchCV	0.6813	0.7154	0.7157	{"C": 0.1, "class_weight": null}
GloVe mean (pretrained)	Logistic Regression	GridSearchCV	0.6813	0.7154	0.7157	{"C": 0.1, "class_weight": null}
GloVe mean (pretrained)	Linear SVC	RandomizedSearchCV	0.6822	0.7168	0.7167	{"C": 0.1, "class_weight": null}
GloVe mean (pretrained)	Linear SVC	GridSearchCV	0.6843	0.7168	0.7167	{"C": 10.0, "class_weight": null}
GloVe mean (pretrained)	RBF SVC	RandomizedSearchCV	0.6684	0.6676	0.6638	{"C": 10.0, "class_weight": "balanced", "gamma": "scale"}
GloVe mean (pretrained)	RBF SVC	GridSearchCV	0.6684	0.6676	0.6638	{"C": 10.0, "class_weight": null, "gamma": "scale"}
GloVe mean (pretrained)	K-Nearest Neighbors	RandomizedSearchCV	0.5895	0.6399	0.6257	{"n_neighbors": 3, "weights": "uniform"}
GloVe mean (pretrained)	K-Nearest Neighbors	GridSearchCV	0.6128	0.6606	0.6409	{"n_neighbors": 7, "weights": "distance"}
GloVe mean (pretrained)	Decision Tree	RandomizedSearchCV	0.5856	0.5893	0.5982	{"criterion": "entropy", "max_depth": 50, "min_samples_leaf": 1}
GloVe mean (pretrained)	Decision Tree	GridSearchCV	0.5527	0.5752	0.5817	{"criterion": "gini", "max_depth": null, "min_samples_leaf": 1}
GloVe mean (pretrained)	Random Forest	RandomizedSearchCV	0.6504	0.6995	0.6932	{"max_depth": null, "max_features": "sqrt", "min_samples_leaf": 2}
GloVe mean (pretrained)	Random Forest	GridSearchCV	0.6321	0.6821	0.6718	{"max_depth": null, "max_features": "sqrt", "min_samples_leaf": 1}
GloVe mean (pretrained)	SGD Classifier	RandomizedSearchCV	0.6545	0.7131	0.7088	{"alpha": 0.001, "class_weight": null, "loss": "log_loss"}
GloVe mean (pretrained)	SGD Classifier	GridSearchCV	0.6538	0.7131	0.7102	{"alpha": 0.001, "class_weight": null, "loss": "hinge"}
GloVe mean (pretrained)	Linear SVC	Optuna (30 prób)	0.6822	0.7168	0.7167	{"C": 0.1, "class_weight": "balanced"}

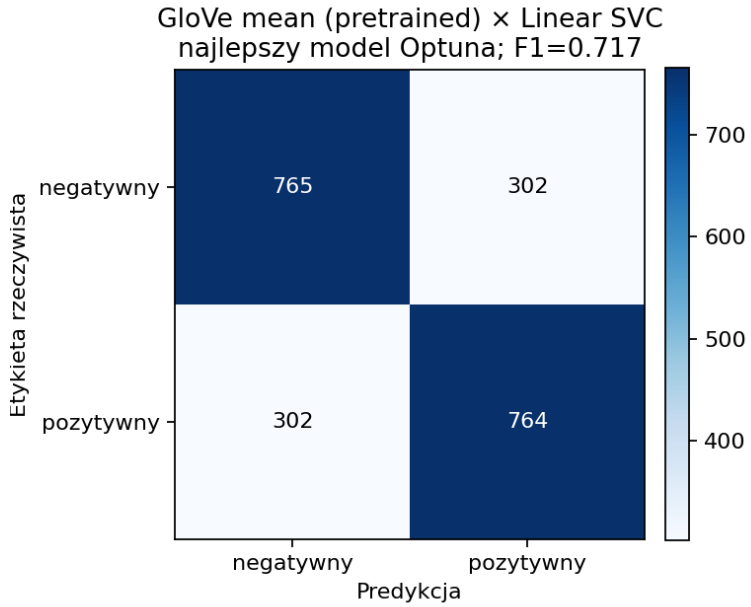


Rys. 1. Wyniki najlepszych konfiguracji według CV F1.



Rys. 2. Testowy F1 dla 42 par reprezentacja–klasyfikator.

6. Macierz pomyłek — model Optuny



Macierze dla 42 par (najlepszy wariant Random/Grid według walidacyjnego F1) zapisano osobno w `Lab4/wykresy/macierze_pomylek/`. Dane TN/FP/FN/TP dla obu metod znajdują się w `Lab4/wyniki/all_results.csv`.

7. Wnioski i ograniczenia

Najlepszy model każdego strojenia w tabeli wybrano według CV F1, a nie wyników testowych. Rezultaty zależą od ograniczonych przestrzeni parametrów i 30-próbkowego budżetu Optuny. Średnie embeddingów pomijają kolejność słów; Word2Vec i Doc2Vec były uczone na niewielkim zbiorze, a GloVe ma niepełne pokrycie słownictwa. Badanie dotyczy angielskich recenzji filmowych i jednego stratyfikowanego holdoutu.

Szczegółowa tabela wszystkich 85 wyników, parametrów i macierzy pomylek znajduje się w `Lab4/wyniki/all_results.csv`; pełny opis implementacji, wszystkich wyników i źródeł w pliku `Sprawozdanie_Lab4.md`.

Środowisko: Python 3.13.2; NLTK 3.9.1; scikit-learn 1.6.1; Gensim 4.4.0; Optuna 5.0.0; Matplotlib 3.10.1.